PCIR: An Open-World Remote Sensing Image Representation Learning Method From a Causal Perspective

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***Abstract*—In the open world, visual representations of remote sensing images vary significantly and diversely. A model can correctly recognize and process unknown data in changing and open-ended scenarios only by learning invariant visual representations from the data. Causal diagram models, which describe the data generation process, can employ interventions and counterfactual operations to obtain invariant visual representations in such open environments, i.e., generic causal representations. However, these generic causal diagram models often overlook the multifeature, multiconfounding, and multichange characteristics of remote sensing images, as well as the causal differences these factors introduce into the imaging process. This limitation makes it difficult for the learned representations to be widely transferable in open scenes. Specifically, we argue that different objects are influenced by different environments, and the styles they exhibit in remote sensing images consequently vary. Therefore, the causal diagrams corresponding to these objects also differ. Intervening at the object-level causal diagram is crucial for obtaining object-specific causal representations and mitigating the problem of feature confusion. Based on this assumption, we derive the general form of the object-level causal diagram from the generic causal diagram, taking into account the characteristics of remote sensing images with multiple features and multiple confounders. We propose a remote sensing representation learning method, PCIR, which is based on a causal partition intervention (CPI) strategy. The method consists of two main components: the CPI module and the causal representation separation (CRS) module: 1) *CPI module:* to simulate the style differences caused by environmental influences on different objects, we propose remote sensing offset data augmentation. We then use target detection labels to identify independent object regions in the image and apply feature-specific data augmentation and 2) *CRS module:* for the same class, the content remains consistent across different styles; therefore, we propose a symmetric causal regularity learning strategy to bidirectionally constrain and obtain invariant causal representations. We validate our method on four datasets across two tasks. The experimental results demonstrate that PCIR achieves outstanding performance in open-scenario tasks. Compared with seven baseline methods, our method improves overall accuracy (OA) by an average of 1.79% in the scene classification task and increases mean accuracy (mAcc) by up to 3.55% in the semantic segmentation task. Moreover, visualizations of the learned representations show that PCIR enhances the model's ability to recognize invariant causal representations in complex scenes and improves its robustness.**

***Index Terms*—Causal inference (CI), remote sensing image, representation learning.**

I. Introduction

In THE open world, there are diverse biases in image datasets due to a variety of factors, such as acquisition methods and environmental changes, and it has been shown that if a model trained on the source data domain is directly applied to differently distributed target data, catastrophic performance degradation usually occurs [1]. To obtain a representational learning ability that can be transferred across different datasets, invariant representation learning research has been proposed [2], [3], [4], [5]. However, we find that the representation learning methods developed for natural images fail to achieve good results when directly transferred to remote sensing images. The core problem lies in the fact that the diversity of scenes and feature types in remote sensing images [6] is much greater than that in natural images, which makes representation learning for remote sensing images more challenging.

In recent years, a series of studies and explorations has been carried out on remote sensing image representation learning [7], [8], [9], [10], [11], [12], [13], [14], [15], [16] to develop interpretation methods that can adapt to remote sensing changes. These methods can be categorized into three main types: manual feature-driven, data feature-driven, and causal feature-driven. Before the emergence of deep learning, the features used in supervised learning were mostly simple spectral features, texture features, and manually designed handcrafted features, which are common features such as histogram of oriented gradient (HOG) [7] and scale-invariant feature transform (SIFT) [8]. With the popularization of high-performance computing, data-driven methods represented by deep learning [9], [10], [11], [12], [13] have been widely developed, especially self-supervised learning [10], [11], [12], [13], which eliminates the dependence on large-scale labeled remote sensing data. In addition, the paradigm of self-supervised models "obtaining common features + fine-tuning" has greatly improved the generalizability of the representations and the transferability of the models. With the increasing scale of remote sensing data, there are increasingly more biases in the dataset, which makes

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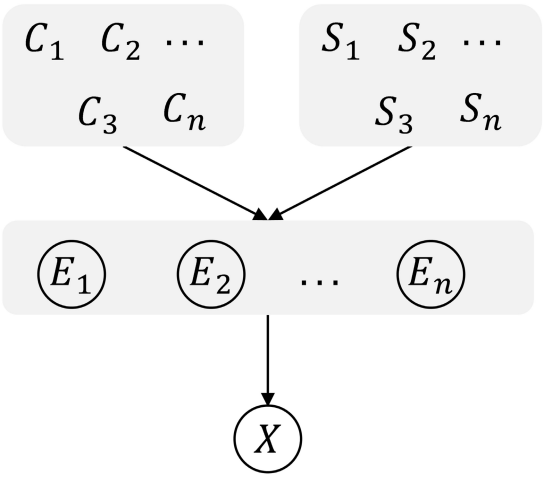


Fig. 1. Relationship between ground objects and image composition in remote sensing images.

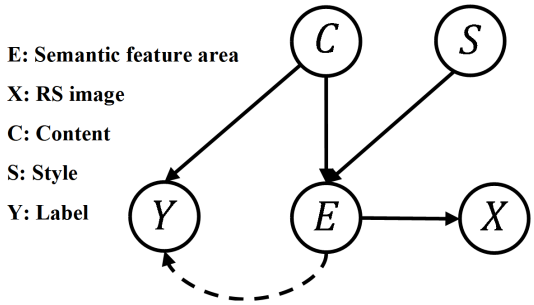


Fig. 2. Causal graph model derived based on our assumptions that can characterize the generation mechanism of remote sensing images.

fine-tuning increasingly difficult. Furthermore, it has been reported that it is difficult for data-driven deep learning models to learn truly invariant representations due to the dependence on statistical correlations between data [17], [18]. For example, for image classification or target detection tasks, airplanes and airports often appear as highly correlated features, but when an airplane is parked on a green lawn, the model may make an incorrect prediction; this is because the model uses the background of the airport as one of the characteristics to identify the airplane, whereas the intrinsic properties of the airplane itself—such as its shape, are the fundamental determinants of "an airplane is an airplane."

To overcome the limitations of data correlations, causal inference (CI) [24] has been introduced into representation learning. Causality reveals the underlying causal mechanisms in the process of data generation, and CI captures stable content within data by performing interventions or counterfactual reasoning on causal relationships to migrate and generalize across different tasks and environments [18]. This stable content, which is referred to as causal features, has attracted increasing attention in areas such as visual understanding and explainable learning [19], [20], [21], [22], [23]. In the field of remote sensing, several studies [14], [15], [16] have introduced CI into the intelligent interpretation of images. Most of these approaches, such as natural images, use global intervention and counterfactual operations to obtain generic causal representations under independent causal mechanisms or other CI methods, but they overlook the characteristics of remote sensing images with multiple features, complex mixtures, and diverse variations. As a result, the characteristics of different ground objects are confused, preventing the model from learning personalized representations of a single ground object and relies only on fuzzy global features for domain adaptation.

We observe that different objects may present the same style under different environments, while the same objects may also present different styles, e.g., eutrophic water bodies and grasslands may both appear uniformly green in remote sensing images, and croplands in different seasons may also present completely different styles, as shown in Fig. 1. In this case, the unified global causal intervention causes the causal representations learned by the model to interfere with and confuse one another, thus losing the detailed description of each object and reducing the model's ability to distinguish between them. Therefore, we believe that the determination of causal representations and styles is relative, and we cannot simply assume that the causal representations across an entire remote sensing image are identical; instead, we should look for a method that can learn causal representations independently for different objects in the remote sensing image. For this purpose, we further refine the representation targets into semantic-level object regions, attempt to construct the causal relationships involved in the process of remote sensing image generation at the semantic level, and build a causal diagram model, as shown in Fig. 2.

Specifically, we use the structural causal model (SCM) [24] to formalize the remote sensing image generation process. We define the features that correspond to the same category across different data distributions as content C, such as the shapes of objects like airplanes, vehicles, and ships, and the rest of the features change differently in the open scene and present different expressions, which we call styles S, such as the area or background where an airplane is parked, the surrounding environment, or the color of leaves. The unlabeled observational data X obtained in the open data scenario are composed of individual feature regions, which are composed of content C and style S, and $y = \{Y\}_{t=1}^{T}$ is the data label, which uniquely corresponds to the data category, so the data label is determined by the content only and is not affected by the style.

We use *do calculus* [24] in CI to separate the content from the stylistic representation and then obtain these decisive causal representations. Due to the unobservability of content and style, we achieve the separation of content and style representations indirectly through two operations: the causal partition intervention (CPI) module and the causal representation separation (CRS) module. First, we use the labels of the target detection data to classify different object regions and apply object-specific data augmentation according to the object characteristics to simulate the external disturbances that may occur during the imaging process of remote sensing images under natural conditions, such as atmospheric radiation or sensor aberrations, thereby realizing style diversification under the condition of content invariance. For the images after the intervention, the content representations correspond to the same labels and share similar features, such as shapes, whereas the style representations show obvious differences

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under the intervention operation. We adopt contrastive learning to achieve invariant causal representation learning by drawing the content-similar representations closer and pushing the style-differentiated representations farther. However, the unidirectionality of the metrics of existing methods can lead to causal imbalance, so we propose symmetric causal regularization to enable the model to learn invariant causal representations that are symmetrically consistent in both directions across positive samples so that the model is not overly biased toward any particular environmental distribution but instead compares differences between representations in a balanced manner, thereby better preserving causal invariance.

Experiments demonstrate that our method outperforms the baseline methods across all four datasets and three metrics for both scene classification and semantic segmentation tasks, with an average overall accuracy (OA) improvement of 1.79% in the scene classification task, an improvement in mean intersection over union (mIoU) of up to 1.89%, and an improvement of up to 3.55% in mean accuracy (mAcc) in the semantic segmentation task. Moreover, we conduct ablation experiments on each module of the method, and the results show that each of our proposed modules can be used individually to enhance the learning effect.

In summary, the main contributions of the work in this article are as follows.

1) We observe and obtain the relationship between semantic-level regions and object representations of remote sensing images and propose a semantic-level causal diagram that characterizes the process of remote sensing image generation, further clarifying the causal relationships in remote sensing images.
2) Based on the assumption of a semantic-level causal diagram, we propose a generalized causal partitioning intervention strategy that simulates the style bias of remotely sensed imagery through data augmentation to achieve intervention in semantic-level feature regions and introduces a symmetric causal regularization goal to achieve the aggregation of content representations under different styles.
3) We conduct experiments on two tasks, four datasets, including NWPU-RESISC45, AID, LoveDA, and Potsdam, and the experiments demonstrate that our method outperforms the baselines in all three metrics. We have an average 1.79% improvement in OA in both scene classification tasks and up to 3.55% improvement in mAcc in semantic segmentation tasks.
4) By deeply exploring the causal relationships in remote sensing images, the model's ability to recognize invariant causal representations in complex scenes is further improved, while enhancing robustness and generalization across diverse remote sensing scenarios. This provides a new perspective and practical approach to the representation learning of remote sensing images from the perspective of causality.

II. Related Work

With the development of deep learning, remote sensing image representation learning has roughly experienced three driving stages: artificial feature-driven, data feature-driven, and causal feature-driven. Artificial feature-driven methods rely on manually designed low-level or intermediate visual features (such as HOG [7] and SIFT [8]), which improve the efficiency of understanding and interpretation, but show weak robustness in complex scenes. Data-driven methods, which automatically learn hierarchical and abstract features through deep learning (such as convolutional neural networks (CNNs) [28]), significantly improve classification accuracy, but are still limited by labeled data and data biases. To address these limitations, causal feature-driven methods introduce causal reasoning, which improves the generalization ability of the model by eliminating spurious correlations and capturing stable causal relationships, and becomes a new direction of remote sensing representation learning.

A. Artificial Feature-Driven Representation Learning

Before the emergence of deep learning methods, most of the features commonly used were lower level visual features, including simple spectral features, texture features, and many carefully designed artificial features. Widely used artificial features include the HOG [7] and SIFT [8]. In addition, to further reduce the "semantic gap" between low-level visual features and scene semantic features, mid-level feature expression methods have also been rapidly developed in the field of remote sensing image scene interpretation [25], [26], [27], whose main idea is to obtain more robust feature expression by encoding the underlying features to handle variations in scale, rotation, illumination, and other factors. Compared with manual visual interpretation, traditional supervised classification methods based on artificial features largely reduce the workload of interpretation and improve efficiency, but there are still obvious limitations. From the point of view of feature description, handcrafted features are usually low- or mid-level features, which often makes it difficult to accurately describe a complex scene composed of multiple features. From the perspective of feature robustness, the abstraction degree of artificially constructed features is low, so they are easily interfered with by radiometric differences in remote sensing images, imaging noise, and various features and their shadows in images. In conclusion, the performance improvement of scene interpretation is greatly limited due to the lack of robustness and representation ability of artificial constructed features.

B. Data Feature-Driven Representation Learning

Currently, with the popularization of high-performance computing, data-driven methods represented by deep learning have received widespread attention in the intelligent interpretation of remote sensing images because of their powerful feature-learning capabilities. Among them, CNNs [28], especially deep CNNs (DCNNs) [29], possess strong feature extraction abilities and exhibit excellent hierarchical, structured, and abstract representation capabilities [30], so they perform surprisingly well in remote sensing interpretation tasks. However, supervised learning methods always face high labeling costs, which severely limit the amount of training data available.

To reduce the dependence on labeled data, unsupervised learning methods have been proposed. Unsupervised learning obtains representations by mining the intrinsic structural properties of the data without labels, thus offering significant

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cost advantages in remote sensing applications [31], [32], [33], [34]. In recent years, self-supervised learning, as an emerging learning paradigm, has been widely used in the field of remote sensing representation learning [10], [11], [12], [13], [35], [36], [37], [38]. By designing pretraining tasks to generate pseudo-labels, self-supervised learning extracts features from unlabeled data, thus significantly reducing the reliance on manual annotation. Self-supervised learning involves a variety of methods, such as SimCLR [39] and MoCo [40], jigsaw tasks [41], and rotation prediction [42], [43], which perform well in remote sensing classification and segmentation tasks. For example, SimCLR uses different views generated by data augmentation to achieve robust feature learning by minimizing the distance between similar samples [39]. In addition, generative tasks (e.g., image restoration and super-resolution reconstruction) further enhance the model's ability to capture contextual and spectral features.

Although self-supervised learning has shown excellent potential in remote sensing, the increasing scale of remote sensing data has also led to growing dataset biases, fine-tuning is becoming increasingly difficult, and the transferability and interpretability of representation learning are gradually becoming bottlenecks. Moreover, it has been found [17], [18] that data-driven deep learning models have difficulty learning true invariant representations due to the dependence on the statistical correlation between data. In addition, the high computational demands of deep learning further limit its applicability in large-scale remote sensing scenarios [44].

C. Causal Feature-Driven Representation Learning

To overcome the limitations caused by data correlations, CI has been introduced into representation learning. In recent years, the application of CI [18] in visual representation learning has gradually become a popular research topic. CI models the data generation process through an SCM [24] and employs intervention and counterfactual reasoning [24] to eliminate spurious correlations and uncover causal patterns within data, thereby providing an effective solution to the problem of distributional and dataset biases. Peters et al. [45] laid the theoretical foundation for CI, whereas Schölkopf et al. [17] and Schölkopf [46] further extended its integration with machine learning, particularly in representation learning. Data bias remains one of the core challenges in representation learning, as models often capture spurious correlations during the learning process, leading to significant performance degradation in out-of-distribution scenarios. To address this issue, Schölkopf et al. [17] proposed clarifying the causal relationships among variables through causal representation learning to eliminate spurious correlations and enhance the generalization ability of models.

Currently, causal representation learning in vision relies on two main assumptions: adversarial and invariance. Adversarial-based causal representation approaches argue that spurious correlations learned by deep learning models reduce the robustness of the models [18]. Tang et al. [47] proposed a causal regularization method for the estimation of instrumental variables for adversarial defense. Wang et al. [48] designed an adversarial training pipeline to suppress the influence of confounding factors by simulating different causal effects. As inferred by invariant causal mechanism (ICM), causal mechanisms may be invariant across domains, so learning causal invariance may be more advantageous in terms of the robustness of representations [18]. Typically, existing methods establish a causal diagram of the data generation process via the SCM and subsequently acquire stable causal mechanisms with the help of intervention or counterfactual manipulation. For example, Yue et al. [49] decoupled the entangled semantics in images into causal variables, subsequently generated counterfactual samples via a generative adversarial network (GAN), and then utilized these counterfactual samples as latent variables for causal intervention. Mitrovic et al. [19] proposed a causal graph model of the image generation mechanism, established a causal relationship between the image and the self-supervised agent task, and realized an invariant causal representation learning objective for the representation learning via invariant causal mechanisms (RELIC) to address the generalization problem of the self-supervised model.

In recent years, CI has gradually entered the field of remote sensing representation learning [14], [15], [16] to realize its potential. Pérez-Suay and Camps-Valls [14] introduced CI into the field of remote sensing and Earth sciences and successfully applied it to tasks such as feature classification and change detection. Xu et al. [16] proposed a network based on causal feature extraction, which analyzes the texture properties in remotely sensed data to improve the multiclassification performance of modal remote sensing data, and the study emphasized the potential of CI in dealing with complex scenes. However, the multifeature, multihybrid, and multichange characteristics of remote sensing images make traditional global causal modeling methods difficult to adapt. First, the construction of causal diagrams relies on a priori knowledge of the data generation mechanism, and the complexity of remotely sensed imagery makes this process exceptionally difficult [16]. Second, existing causal modeling methods usually assume independence among variables, whereas the generation mechanisms of different objects in a remote sensing scene are often strongly coupled, and unified modeling may lead to feature confusion. For example, when dealing with remote sensing scenarios involving water bodies, forests, and buildings, the differences in environmental influences on different objects make it difficult for unified modeling to accurately capture the core causal features of each object type [14]. With the continuous expansion of remote sensing tasks, many works have also begun to consider how to construct graph models for remote sensing images. Nie et al. [66] introduced a dynamic spatiotemporal graph neural network for the video question–answering task. By constructing a graph structure that evolves over time, they modeled the causal chain of events, which provides inspiration for constructing causal graphs of remote sensing images in complex scenes. Zheng et al. [67] achieved multiscale and multiangle semantic alignment between remote sensing images and text by constructing a coarse-to-fine perspective matching strategy, taking into account both the consistency and style differences of ground object semantics from multiple perspectives. Although these methods have achieved certain success in their respective tasks, they have not focused on discussing how causal relationships in remote sensing images should be modeled. Therefore, we further explore how to

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fully consider the complexity of remote sensing images in causal modeling and propose a more targeted learning method of object-level causal representation to obtain transferable invariant representations in remote sensing images.

III. Method

In this section, we provide a detailed description and elaboration of the model architecture. We first elaborate on the principles of the common cause and the independent causal mechanism in Section III-A, providing a derivation and proof of the reasonableness of treating the content as an invariant visual causal representation, which serves as the theoretical basis and guidance for the method. Based on the theoretical proof, in Section III-B, we further elaborate on the components of the model, presenting the design concepts and implementation details of each strategy.

A. Invariant Representation Learning

In open scenarios, environmental differences such as sensors and external conditions lead to different distributions and characteristics of images formed by the same features, which greatly increases the difficulty of learning invariant representations in intelligent models for remote sensing [6]. Traditional models rely on the statistical correlations between the labeled data and the images to obtain invariant representations, but these statistical correlations may vary across different distributions and features. Principle 1 [50] shows that causal mechanisms can maintain stable judgments of invariant representations without changing under external disturbances.

Principle 1: Common Cause Principal: If two observable quantities X and Y are statistically correlated, there must be a variable Z that causally influences X and Y and can explain all the correlations between them. In other words, given Z, X, and Y are conditionally independent.

Based on the principle of common cause, we can formalize our causal diagram model in Fig. 2 to obtain

$$X := \{E_1, E_2, \ldots, E_n\} \tag{1}$$

$$E := f(C, S), C \parallel S \tag{2}$$

$$Y := h(g(X)) \tag{3}$$

where X is the input complete remote sensing image, E is the semantic-level feature region, and Y represents the corresponding label. Content C is a causal factor that causally affects both, and style S denotes a noncausal factor that causally affects only E. Thus, for any distribution $P(E, Y) \in P$, given that the contents C, E, and Y are conditionally independent, i.e., there exists a data domain-independent $P(Y \mid C)$ mechanism that generalizes under any data domain. We, therefore, argue that such content C can be used as an invariant representation that generalizes well in open dynamic scenarios.

Principle 2: Independent Causal Mechanism Principle [17]: The conditional distribution of the causes of each variable (i.e., its mechanism) is given in such a way that it does not inform or influence the other mechanisms.

According to the independent causal mechanism, under the current causal model, intervening in S does not change the conditional distribution $P(Y \mid C)$. Therefore, $P(Y \mid C)$ remains unchanged under the change of style S, i.e.,

$$p^{\mathrm{do}(S=s_i)}(Y \mid C) = p^{\mathrm{do}(S=s_j)}(Y \mid C) \quad \forall s_i, s_j \in S \tag{4}$$

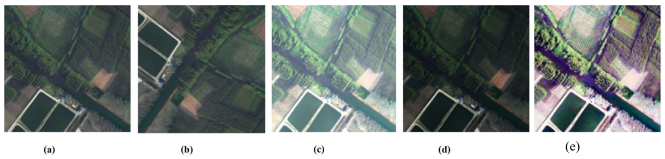


Fig. 3. Use formulas to simulate the possible distortion and offset effects that may occur during various image generation processes. (a) Original images. (b) Rotation, etc. (c) Sensor offset. (d) Radiation offsets. (e) Inherent offset.

where $p^{\mathrm{do}(S=s_i)}$ denotes the distribution generated by assigning the value of s_i to S through an intervention operation. However, during model implementation, we must consider that we cannot realistically access the style S of the images, which is practically determined by the fact that we cannot manipulate the satellites or features during the generation of the remotely sensed images, which means that we cannot intervene in the style S directly. To simulate the change in style, we add content-holding data augmentation to the previously obtained remote sensing image dataset as an intervention operation on the style variable S, i.e.,

$$p^{\mathrm{do}(a_i)}(Y \mid C) = p^{\mathrm{do}(a_j)}(Y \mid C) \quad \forall a_i, a_j \in A. \tag{5}$$

The equivalence derivation of $\mathrm{do}(S = s_i)$ and $\mathrm{do}(a_i)$ is provided in the Appendix. Specifically, as shown in Fig. 3, we analyze the imaging process of remote sensing images and believe that the offset factors that cause changes in the expression of remote sensing images during the imaging process mainly include: 1) intrinsic offset of the target object: climatic changes due to time or seasonal variations; 2) radiometric offset: brightness or spectral distortion caused by the propagation of light conditions, atmospheric effects, etc; and 3) sensor offset: nonideal characteristics of the sensor itself cause imaging aberrations. We subsequently model the three types of offset factors using the formula in the study of [51], which were added to the model as data enhancements. The enhanced effect is shown in Fig. 3

Inherent offset of target features: $X^* = \mathrm{HM}\ (\mathrm{ref}, X)$

Radiation is cheap: $X^* = m \cdot X^{\gamma}$

Sensor Offset: $X^* = m^t \cdot X + m^t$.

Although the variations of remote sensing images in the real world may be more complex—covering structural deformation, seasonal growth changes, and diverse acquisition resolutions—these basic and controllable simulation methods are representative and operable, providing a clear and targeted experimental basis for our initial exploration and verification of causal intervention mechanisms. To effectively verify the effectiveness of the proposed causal intervention theory in learning invariant representations, we design corresponding experiments.

B. Remote Sensing Comparative Learning Approach Based on a Causal Partitioning Intervention Strategy

In this section, we detail the proposed model architecture, whose core design motivation is to address the challenges of causally invariant representation learning in complex distributional scenarios. Previous self-supervised methods [52] mainly focus on the correlations between similar samples

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when learning representations, while ignoring the underlying causal mechanisms within the samples, which may lead to a degradation of the model's generalization ability under different distributions.

Our aim is to conduct semantic-level causal interventions on the ground object regions in remote sensing images, thereby extracting the specific causal features corresponding to each type of ground object. In the causal partitioning intervention module, we divide independent ground object regions based on object detection labels and apply different data augmentations to simulate the style changes of ground objects under different environmental offsets, achieving local intervention on the image. Subsequently, the intervened images are input into the CRS module. Since the content remains unchanged while the style varies, we pull closer the representations with similar content and push apart those with stylistic differences through contrastive learning to learn stable and invariant representations. Considering the problem of metric asymmetry among augmented views in conventional contrastive methods, we introduce causal symmetry regularization to improve representation consistency and enhance causal invariance through bidirectional constraints.

In particular, our module design is universal and compatible with other self-supervised methods, such as ReLIC [19]. Here, we choose the DETR with improved denoising anchor boxes (DINO) method [52] to carry out the main experiments. It is a self-supervised contrastive learning method that uses a transformer as the backbone network, where we set up a teacher network and a student network with the same architecture and respective parameters. The student network is trained by comparing and aligning its output with that of the teacher network, whose parameters are updated using an exponential moving average (EMA), to minimize the difference between them and thus learn the knowledge of the teacher network.

Compared with traditional methods, our model not only focuses on the correlation between similar samples but also significantly enhances the interpretability and robustness of the representations through the modeling of causal mechanisms, providing a brand new perspective and solution for the learning of representations in complex distributional scenarios.

1) Causal Partitioning Intervention Module (CPI): We delineate and segment the object areas containing independent semantics in the image with labels of the object detection dataset and obtain the label area as the object of intervention imposition in the module in Fig. 4(a). To simulate the process in which the remote sensing image is influenced by migration to form a style and the object response to migration presents diverse forms, we combine the remote sensing offset augmentations obtained in Section III-A to form different combinations of data augmentations $\{a_1, a_2, \ldots, a_i\}, a \in \mathcal{A}$, resulting in different expression styles for different categories of objects, which corresponds to the combination hypothesis we propose in Fig. 1.

In actual operation, we will make a judgment based on the label category of the target in the image to determine the type of enhancement to be superimposed on the ground object. To increase the intensity of the offset, we set the thresholds for each enhancement within a certain range based on the actual situation. Within these threshold ranges, we randomly select values and apply them to each enhancement, thereby ensuring the diversity of the enhancements.

We randomly apply the obtained data augmentation combinations to independent semantic object regions to generate diverse style maps and then recombine all object regions into the original remote sensing images, as shown in Fig. 5. On the one hand, these images are input into the teacher network g_{θ_t} as global views, and on the other hand, randomly generated local views are input into the student network g_{θ_s}. Feature extraction is performed in both networks to obtain the outputs $g_{\theta_t}(x)_i^{a_l}$ and $g_{\theta_s}(x)_i^{a_h}$. By normalizing the outputs using the Softmax function, we can obtain the probability distributions of the features across the K dimensions

$$p_t^{\mathrm{do}(a_l)}(x)_i = \frac{\exp\left(g_{\theta_t}(x)_i^{a_l}\right)/\tau}{\sum_{k=1}^{K}\exp\left(g_{\theta_t}(x)_k^{a_l}\right)/\tau} \tag{6}$$

$$p_s^{\mathrm{do}(a_h)}(x)_i = \frac{\exp\left(g_{\theta_s}(x)_i^{a_h}\right)/\tau}{\sum_{k=1}^{K}\exp\left(g_{\theta_s}(x)_k^{a_h}\right)/\tau} \tag{7}$$

where x is the original image, a_l and a_h are different combinations of data enhancement, and do$(\cdot)$ indicates that combinations of data augmentations are imposed on the image. g_{θ_t} is the teacher network used to receive the global view. g_{θ_s} is the student network, which is used to receive the local view for feature extraction of the input image. $g_{\theta_s}(x)_i^{a_l}$ and $g_{\theta_t}(x)_i^{a_l}$ indicate the results of feature extraction of the enhanced input image by the teacher network and the student network. The temperature coefficient τ controls the sharpness of the output distribution, and k is the feature dimension.

2) CRS Module: The encoded features are usually confounded, and to obtain stable and robust representations, we need to achieve the separation and extraction of causal representations. Contrastive loss, as the core of contrastive learning, enables invariant representation learning by increasing the similarity of positive samples and decreasing the similarity of negative samples. The commonly used loss objective, such as InfoNCE [53], can be written as

$$\mathcal{L}_{\mathrm{Info}} = -\log \frac{\exp\left(\mathrm{sim}\left(g_{\theta_t}(x)_i^{a_l}, g_{\theta_s}(x)_i^{a_h}\right)/\tau\right)}{\sum_{m=0}^{M}\exp\left(\mathrm{sim}\left(g_{\theta_t}(x)_i^{a_l}, g_{\theta_s}(x)_m^{a_h}\right)/\tau\right)}. \tag{8}$$

To further constrain invariant prediction, the use of kullback-leibler (KL) divergence as an explicit regularizer to enforce invariance in the loss targets has been proposed [19]

$$\mathcal{L}_b \text{ s.t. } \mathrm{KL}\left(p_t^{\mathrm{do}(a_l)}(x), p_s^{\mathrm{do}(a_h)}(x)\right) < \rho. \tag{9}$$

Although intended to constrain the distance between similar samples, we observed that the distribution distance between samples from two different enhanced views was asymmetric, i.e., $\mathrm{KL}(a_i, a_j) \neq \mathrm{KL}(a_j, a_i)$. This asymmetry contradicts our causal assumption because we assume that representations with the same labels should not be affected by confounding factors. The jensen-shannon (JS) divergence [54] provides a more balanced way to measure and minimize the differences in distributions across different environments, helping to ensure that the model does not overly favor any specific environmental distribution and thereby better promotes causal invariance.

To this end, we propose a symmetric causal regularization objective, enabling the model to learn bidirectionally

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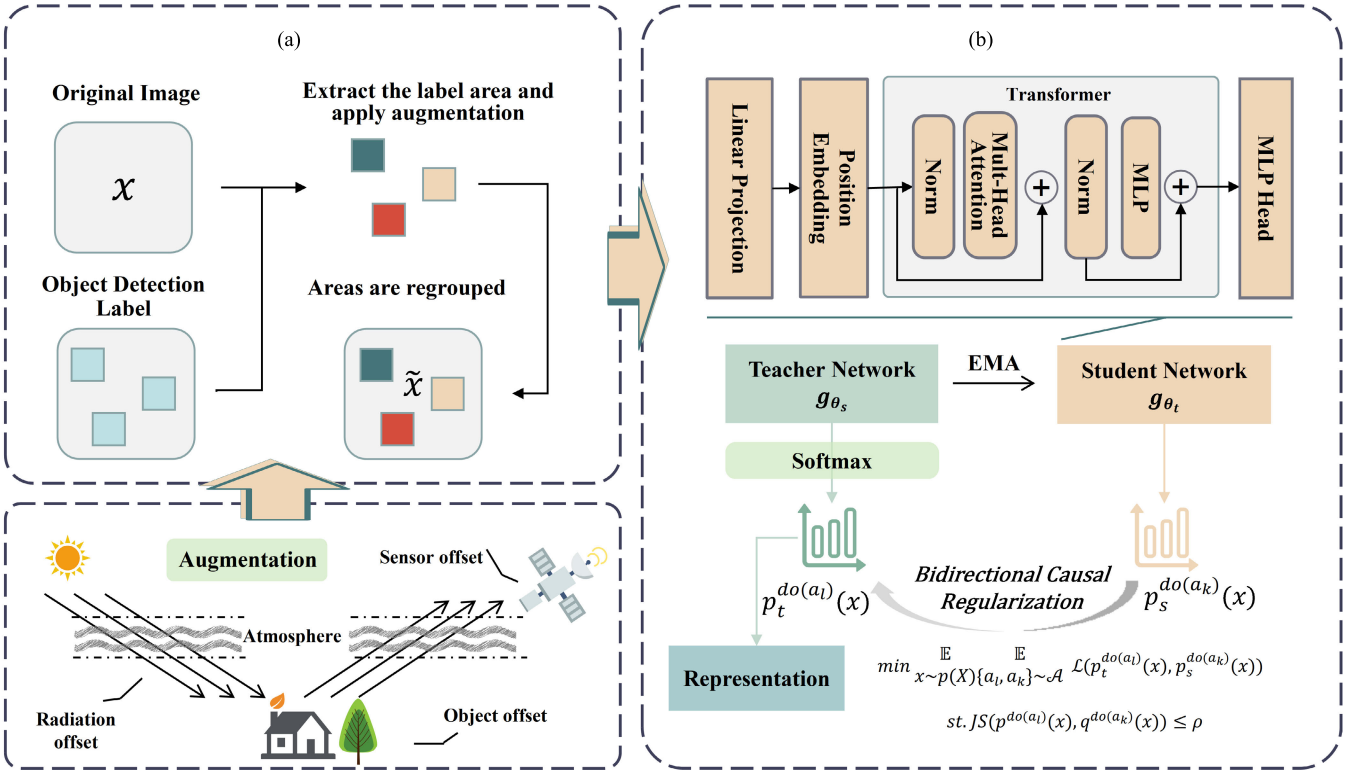


Fig. 4. Model framework. (a) Causal partitioning intervention module. This module receives the original image and the target detection label, extracts and regroups the label area, and generates the enhanced image. The enhancement process takes into account a variety of physical factors, such as radiation offset, sensor offset, and target offset. (b) CRS module. This module contains a Transformer-based structure for learning image representations. Through the teacher network and the student network, combined with bidirectional causal regularization, the aim is to separate the causal representations that are domain-invariant and domain-related, in order to improve the generalization ability of the model.

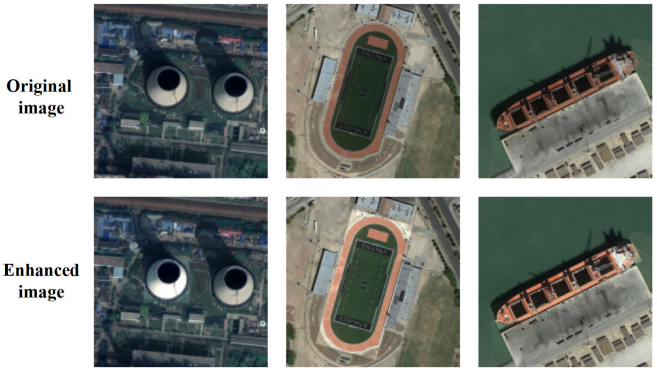


Fig. 5. Partition intervention simulated the offset in the generation process of real remote sensing images. After partitioning with the help of target detection labels and adding offset simulation, a complete image with diverse presentation forms can be obtained after combination.

symmetric and consistent causal representations among positive samples. The target formula is as follows:

$$\min \mathop{\mathbb{E}}_{\{a_l, a_h\} \sim \mathcal{A}} \mathcal{L}\left(g_{\theta_t}(x)_i^{a_l}, g_{\theta_s}(x)_i^{a_h}\right)$$
$$\text{st. JS}\left(p_t^{\text{do}(a_l)}(x), p_t^{\text{do}(a_h)}(x)\right) < \rho \tag{10}$$

where $\mathcal{L}(\cdot)$ represents the similarity measure between the outputs of the student network and the teacher network, and JS$(\cdot)$ represents the JS divergence, which treats the two distributions equally and does not specify a reference. We use a weighting factor to control the strength of the regularization term, which can be adjusted as needed.

Combining the above analysis, we achieve our invariant representation learning by minimizing the following goals:

$$-\sum_{i=1}^{N}\sum_{a_{lh}} \log \frac{\exp\left(\text{sim}\left(g_{\theta_t}(x)_i^{a_l}, g_{\theta_s}(x)_i^{a_h}\right)/\tau\right)}{\sum_{m=1}^{M}\exp\left(\text{sim}\left(g_{\theta_t}(x)_i^{a_l}, g_{\theta_s}(x)_m^{a_h}\right)/\tau\right)}$$
$$+\alpha\sum_{a_{lh}} \text{JS}\left(p^{\text{do}(a_l)}(x), p^{\text{do}(a_h)}(x)\right) \tag{11}$$

where $x_i^{a_l}$ and $x_i^{a_h}$ are the images obtained after combining a_l and a_h for data augmentations, respectively, M is the number of labeling categories, sim$(\cdot)$ represents the difference between the images after different data augmentation, and τ is the temperature coefficient. This objective forces the internal representations of the images (even if their appearances differ) to gradually aggregate toward invariance while suppressing the changing style representations. This ensures the robustness of the model at the content representation level. Furthermore, this regularization imposes bidirectional constraints on the distributions of the two images to avoid excessive bias of the model toward the distribution of a specific environment, thereby better capturing causal invariance on the basis of a balanced comparison of differences.

IV. Experiment and Analysis

A. Experimental Setup

We evaluate the performance of our proposed PCIR method through four experiments.

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1) One of the important and intuitive indicators of whether invariant representations are learned is the model's generalization ability in open scenarios. Therefore, we select four representative datasets under two different task settings for evaluation. The results verify that our model demonstrates better generalization and transferability than the baseline models across different data domains and task scenarios.
2) To more intuitively illustrate how much the model focuses on the invariant representation regions, we visualize the self-attention mechanism of the [CLS] token.
3) We evaluate the two modules of the model—causal partitioning intervention and CRS—to verify the contribution of each component to overall performance improvement.
4) To demonstrate that the proposed strategy combination is a relatively independent structure applicable to other self-supervised methods, we apply it to different self-supervised frameworks and compare the results with their original versions to verify its transferability.

All the experiments adopt the pretraining–fine-tuning paradigm for model training and testing; that is, after training on the pretraining dataset, the weights are transferred to different datasets for scene classification and semantic segmentation. The specific experimental settings are as follows.

1) Dataset: To ensure the credibility of the experimental results, we select public and widely used datasets as research objects in our experiments.

In the pretraining phase, we need to perform partitioning interventions with the help of target detection labeling information; therefore, we choose an object detection dataset with comprehensive labeling information and a large data volume. DIOR [55] is a dataset widely used for target detection in remote sensing image analysis, containing 23 463 high-resolution remote sensing images of 800 × 800 pixels, with target objects covering 20 different categories. In total, it includes more than 192 000 target instances. The images encompass common geographic targets and man-made structures such as airplanes, airports, boats, vehicles, golf courses, and playgrounds. The DIOR dataset provides precise bounding-box annotations for each target, and we utilize its rotated target-box annotations to perform causal partitioning interventions; however, during pretraining, we still use the unlabeled dataset.

In the downstream stage, we select two classical tasks of remote sensing image interpretation, namely, scene classification and semantic segmentation. Specifically, we use four datasets—NWPU-RESISC45 [56], AID [57], LoveDA [58], and Potsdam [59]—as the final datasets for test.

NWPU-RESISC45 [56] is a widely used remote sensing image scene classification dataset containing 31 500 remote sensing images with a resolution of 256 × 256 pixels. It includes 45 different types of scenes, such as airports, baseball stadiums, business districts, forests, parking lots, harbors, railway stations, rivers, runways, storage tanks, and tennis courts, with 700 images for each scene category. These images exhibit substantial diversity, including varying lighting conditions, seasonal changes, and complex backgrounds, making NWPU-RESISC45 an important benchmark dataset for evaluating remote sensing scene classification algorithms.

AID [57] is a large-scale dataset for scene classification in remote sensing images. The AID dataset contains a total of 10 000 remote sensing images with a resolution of 600 × 600 pixels, covering 30 different scene categories such as airports, beaches, bridges, business districts, forests, farmlands, industrial areas, lakes, residential areas, rivers, sports fields, and parking lots. Each category includes approximately 220–420 images. The AID dataset poses a significant challenge to classification algorithms due to the diverse and complex scene categories in the images, as well as the substantial variations that may exist within the same category (e.g., different seasons, weather conditions, etc.).

LoveDA [58] is a land-cover remote sensing dataset designed for semantic segmentation tasks. It contains 5987 high-resolution remote sensing images with a size of 1024 × 1024 pixels, categorized into two main types of scenes: urban and rural. The dataset covers seven primary categories, namely, buildings, roads, vegetation, water, barren, forest, and background, and provides accurate pixel-level annotations. The diversity between urban and rural scenes in the dataset places high demands on the generalization capability of the algorithm.

Potsdam [59] is a high-resolution remote sensing dataset designed for semantic segmentation tasks, particularly in urban environments. It contains 38 aerial images, each with a resolution of 6000 × 6000 pixels and a ground sampling distance (GSD) of 5 cm. The dataset covers a wide range of urban and suburban scenes and is labeled with categories such as buildings, impermeable surfaces (e.g., roads), trees, vehicles, and clutter/background. It provides accurate pixel-level annotations, making Potsdam an important benchmark for evaluating an algorithm's generalization capability and segmentation accuracy in high-resolution urban landscapes. The rich diversity of urban scenes in the dataset poses significant challenges to the robustness and performance of algorithms.

2) Baseline: We selected seven state-of-the-art methods, such as SimCLR [39], BYOL [60], MoCo v2 [40], DINO [52], ReLIC [19], Grass [13], and LCR [61], as baselines for evaluating our model from the perspectives of both self-supervised comparison learning and invariant representation learning.

3) Metrics: In the scene classification task, we choose OA as the evaluation metric, which can be defined as

$$\mathrm{OA} = \frac{N_{\mathrm{correct}}}{N_{\mathrm{total}}} \tag{12}$$

where N_{correct} represents the number of individuals correctly predicted and N_{total} represents the overall number of predictions.

The metrics we compare in the semantic segmentation task are the average number of crosslinks (mIoU) and the average class accuracy (mAcc), which are common metrics for semantic segmentation tasks. The definition of the mIoU is given by the following equation:

$$\mathrm{mIoU} = \frac{1}{n}\sum_{i=1}^{n}\frac{U_i^{\mathrm{prediction}} \cap U_i^{\mathrm{target}}}{U_i^{\mathrm{prediction}} \cup U_i^{\mathrm{target}}} \tag{13}$$

where n represents the number of categories, $U_i^{\mathrm{prediction}}$ represents the predicted area of feature category i, and U_i^{target} represents the true area of feature category i.

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TABLE I
COMPARISON OF THE QUANTITATIVE RESULTS BETWEEN THE BASELINE METHOD AND PCIR IN THE SCENE CATEGORIZATION TASK

Data/Task	Pretain Data:DIOR	
	NWPU-RESISC45 (OA)	AID (OA)
SimCLR	75.89	73.95
BYOL	76.03	73.44
MoCo v2	76.35	75.98
DINO	77.04	76.75
ReLIC	74.79	74.40
Grass	77.21	76.87
LCR	76.29	76.75
(PCIR)Ours	77.50	77.76

The definition of the mAcc can be expressed as

$$\text{mAcc} = \frac{1}{n}\sum_{i=1}^{n}\frac{P_i^{\text{correct}}}{P_i^{\text{total}}} \tag{14}$$

where n represents the number of categories, P_i^{correct} represents the number of pixels correctly predicted for the feature i, and P_i^{correct} represents the number of pixels predicted for the feature i.

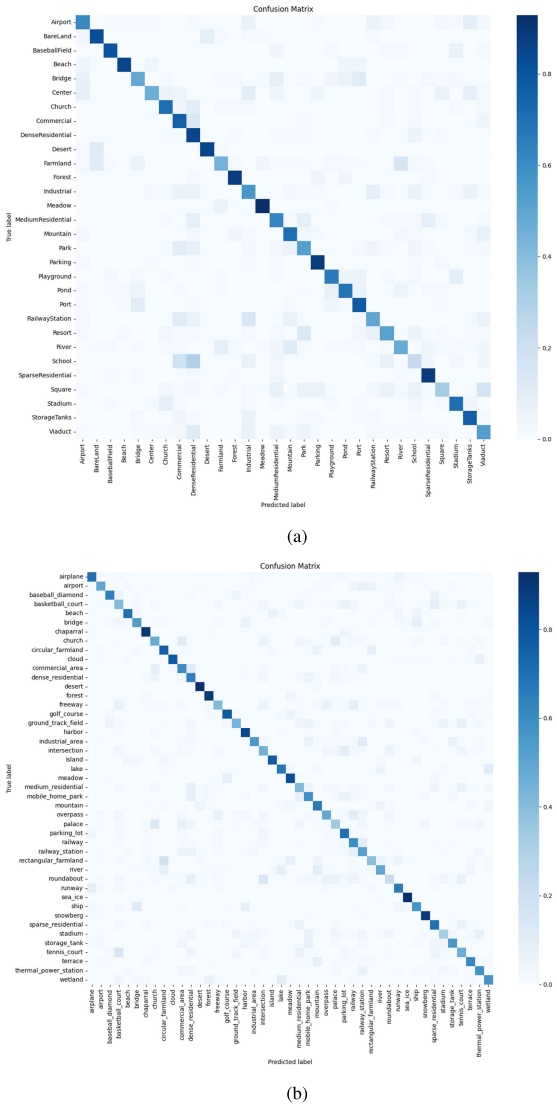
4) Implementation Details: In the self-supervised pretraining stage, we train for 300 epochs using the entire training dataset without labels, with the batch size set to 32. For each baseline method, we adopt the data augmentation and optimization settings recommended in the original article, and all self-supervised pretraining methods are used solely to train the feature extractor.

In the downstream fine-tuning phase of remote sensing image (RSI), to accurately evaluate the performance of features extracted by different self-supervised pretraining methods, we freeze the weights of the pretrained backbone network and update only the parameters of the feature decoder used to obtain RSI classification and segmentation results. For the scene classification task, we use a linear evaluation protocol, in which the backbone network is initialized with the pretrained weights and fixed as the feature extractor, while only the linear classifier weights are trained. The dataset is divided into 80% for training and 20% for testing to evaluate the quality of the pretrained features. For the semantic segmentation task, we transfer the pretrained weights to the *mmsegmentation* framework for testing and adopt Deeplabv3 + (based on ResNet) and SegFormer (based on Transformer) as the segmentation models. Each model is trained for 80k iterations according to its backbone, and all other settings follow the standard configuration of *mmseg*.

B. Results

1) Verification of the Generalization Ability: In order to verify the generalization capability of our method, we test two tasks of scene classification and semantic segmentation, respectively.

a) Scene Categorization Tasks: For the scene classification task, we compare the PCIR with the seven baseline methods, and the experimental results are shown in Table I. The results show that, compared with the remaining seven baseline methods, PCIR achieves the best results in terms of OA metrics on both the NWPU-RESISC45 and AID datasets. For the NWPU-RESISC45 dataset, our method slightly outperforms Grass and MoCo v2, but there is more than a 1% improvement in effectiveness compared with all the other methods, with the greatest improvement of 2.71%. However, on the AID dataset, our method has more than a 1% improvement over all baseline methods, with a maximum improvement of 4.32%, which indicates that our method has a stable effect improvement on large-scale datasets and better classification results for datasets with more categories and larger sizes.



(a)

(b)

Fig. 6. Prediction confusion matrix is experimentally obtained on the (a) NWPU-RESISC45 dataset and (b) AID dataset.

Furthermore, we analyze the recognition accuracy of the model for each category, as shown in Fig. 6. The confusion matrix demonstrates that our model achieves strong performance in most categories, particularly in those with clearer boundaries, such as "Beach," "Meadow," "Parking," and "StorageTanks," as well as other categories with more distinct visual separations. We believe that our intervention on individual semantic regions—by delineating causal regions—enables the model to more clearly and effectively learn and recognize the semantic boundaries of each category,

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TABLE II

COMPARISON OF THE QUANTITATIVE RESULTS BETWEEN THE BASELINE METHODS AND PCIR IN SEMANTIC SEGMENTATION TASKS

Data/Task	Pretrain Data:DIOR			
	Potsdam (mIoU,mAcc)		LoveDA (mIoU,mAcc)	
SimCLR	43.02	54.90	33.09	45.31
BYOL	43.93	55.73	37.39	48.34
MoCo v2	42.81	54.53	36.92	45.77
DINO	44.03	56.71	34.77	45.24
ReLIC	42.70	53.86	35.47	46.92
Grass	44.39	56.80	36.77	47.77
LCR	43.88	55.52	36.53	47.97
(PCIR)Ours	44.87	57.41	37.44	48.79

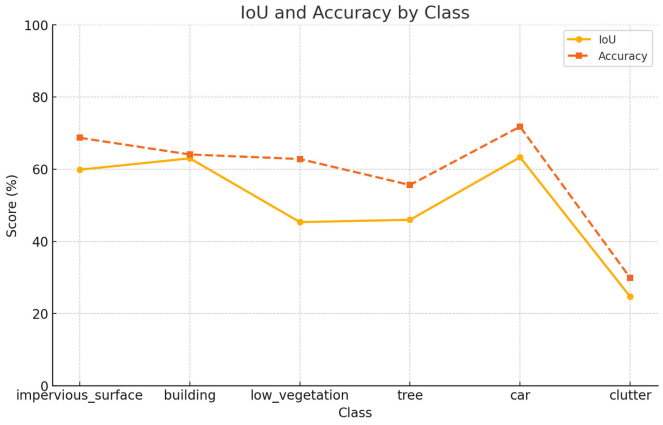
thereby focusing more precisely on the core causal features. However, we also observe that although our method effectively mitigates confusion and spurious correlations caused by style, its ability to address the inherent complexity of the content itself remains limited. For instance, misclassifications occur in highly confusable categories such as "Dense Residential" and "Medium Residential," as well as between "Wasteland" and "Forest." This may be partly because these categories share highly similar characteristics, while our model has not yet achieved fine-grained feature recognition within each semantic region. Moreover, the intrinsic complexity of the "content" itself often leads to confusion, making it more difficult for the model to accurately disentangle content from style. In future work, we plan to enhance the model's capacity to capture differences at a finer granularity.

b) Semantic Segmentation Task: For the semantic segmentation task, we evaluated all methods on two datasets, Potsdam and LoveDA, and the experimental results are presented in Table II. The results show that, compared with other baseline methods, our approach achieves the best performance on both metrics across both datasets. Our method demonstrates a more stable and consistent improvement in mIoU, with an average gain of 1.33% on the Potsdam dataset and 1.59% on LoveDA. The improvement in mAcc is even more pronounced, reaching 1.97% and 2.03% on the two datasets, respectively, with the maximum improvement being 3.55%. These results indicate that our method achieves the most stable and significant improvements in semantic segmentation performance on different data domains. In addition, we observe that the improvement in segmentation accuracy achieved by our method on the Potsdam dataset is slightly lower than that on LoveDA. This may be because most of the features in the urban scenes of the Potsdam dataset exhibit greater similarity and more complex boundaries, posing a greater challenge for the model in obtaining effective causal representations.

We visualize the IoU and accuracy for each feature category in the LoveDA and Potsdam datasets, as shown in Fig. 7. The experimental results show that PCIR performs well on the water and building categories but exhibits poorer segmentation performance on categories such as wasteland and forest, which have relatively low IoUs and accuracies. Furthermore, we visualize the segmentation results, as shown in Fig. 8. From the segmentation maps, we find that features such as vehicles, water bodies, and buildings have more regular boundaries in the remotely sensed images. As a result, we



(a)



(b)

Fig. 7. IoU and Acc for each category of the dataset. (a) LoveDA dataset for seven categories of IoU and Acc. (b) Potsdam dataset for five categories of IoU and Acc.

can obtain more complete content regions during partitioning, providing a better intervention effect for subsequent invariant representation extraction. In contrast, for categories such as wasteland, their boundaries are more blurred, and they are less responsive to the modeled bias factors, leading to less obvious intervention effects. To further improve model performance, we plan to explore more effective operational strategies for partitioned feature intervention in future work, with the aim of enhancing the model's recognition of these categories and improving overall segmentation accuracy.

2) Attention Visualization Results: To further visualize whether our model indeed learns the invariant representation regions as expected, we visualize the neural network in our model. In the scene classification task, by visualizing the self-attention mechanism of the [CLS] token [62], we can intuitively assess the model's attention to the input sequence across different attention heads and determine whether it effectively captures the key features in the input. Specifically, we extract the final self-attention layer in the vision transformer (ViT)-small architecture of the model's backbone network and analyze the attention regions of each attention head. The results are shown in Fig. 9, where each subfigure displays the input image and its corresponding visualization under different attention heads. Different colors in the visualization

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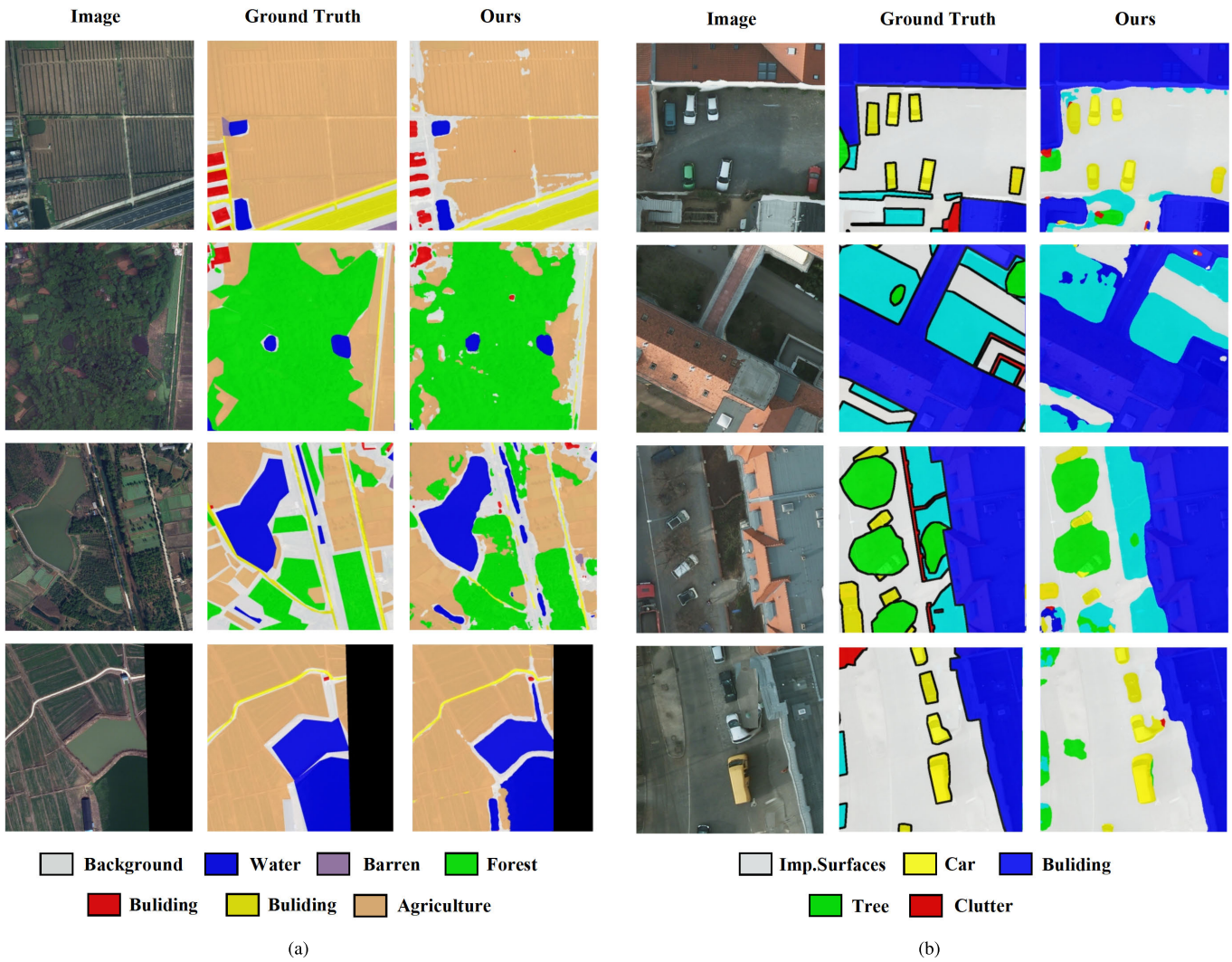


Fig. 8. Comparison of the segmentation results between ours and ground truth. (a) LoveDA. (b) Potsdam.

indicate the strength of the attention distribution. The results demonstrate the following.

1) Our model can focus its attention on the core feature regions in the image, such as buildings, airplanes, ships, and other objects with distinctive characteristics. The attention regions differ slightly across the various attention heads, indicating that the model attends to the input features in diverse ways. This suggests that the model exhibits a certain degree of robustness in capturing features at different scales and levels of detail.
2) The attention regions of the model are largely consistent with the main content perceived by humans, indicating that the model focuses its attention on the representative and stable areas of the image. This consistency demonstrates that the model effectively captures stable and semantically meaningful invariant representations in the input images during the learning process. The experimental results verify that the model can identify important regions and invariant representations in the input, thereby achieving the learning objective of extracting stable and invariant features from the input images. These findings provide strong evidence for the capability of our proposed model to capture key features in visual tasks.

3) Ablation Experiments: To explore the separate validity of each module in our proposed approach, we conducted ablation experiments. Our method mainly includes two modules: CPI and CRS. We believe that simulating remote sensing image migration also plays an important role, so we conducted ablation experiments on the remote sensing migration simulation enhancement of the two modules and the CPI module in this section. The experimental results are shown in Table III.

Specifically, four groups of experiments were conducted, with one dataset selected for each of the two tasks for testing. The first group does not incorporate any of our proposed strategies, i.e., it uses the original baseline method. The second group adds only the CRS module. The third group introduces the CPI module but applies standard image data augmentation. The fourth group incorporates the CPI module along with the data augmentation method that simulates remote sensing image offsets. The results show that all of our strategies have a positive enhancement effect on downstream image understanding tasks. Among them, the CRS module provides the most significant improvement across the three metrics

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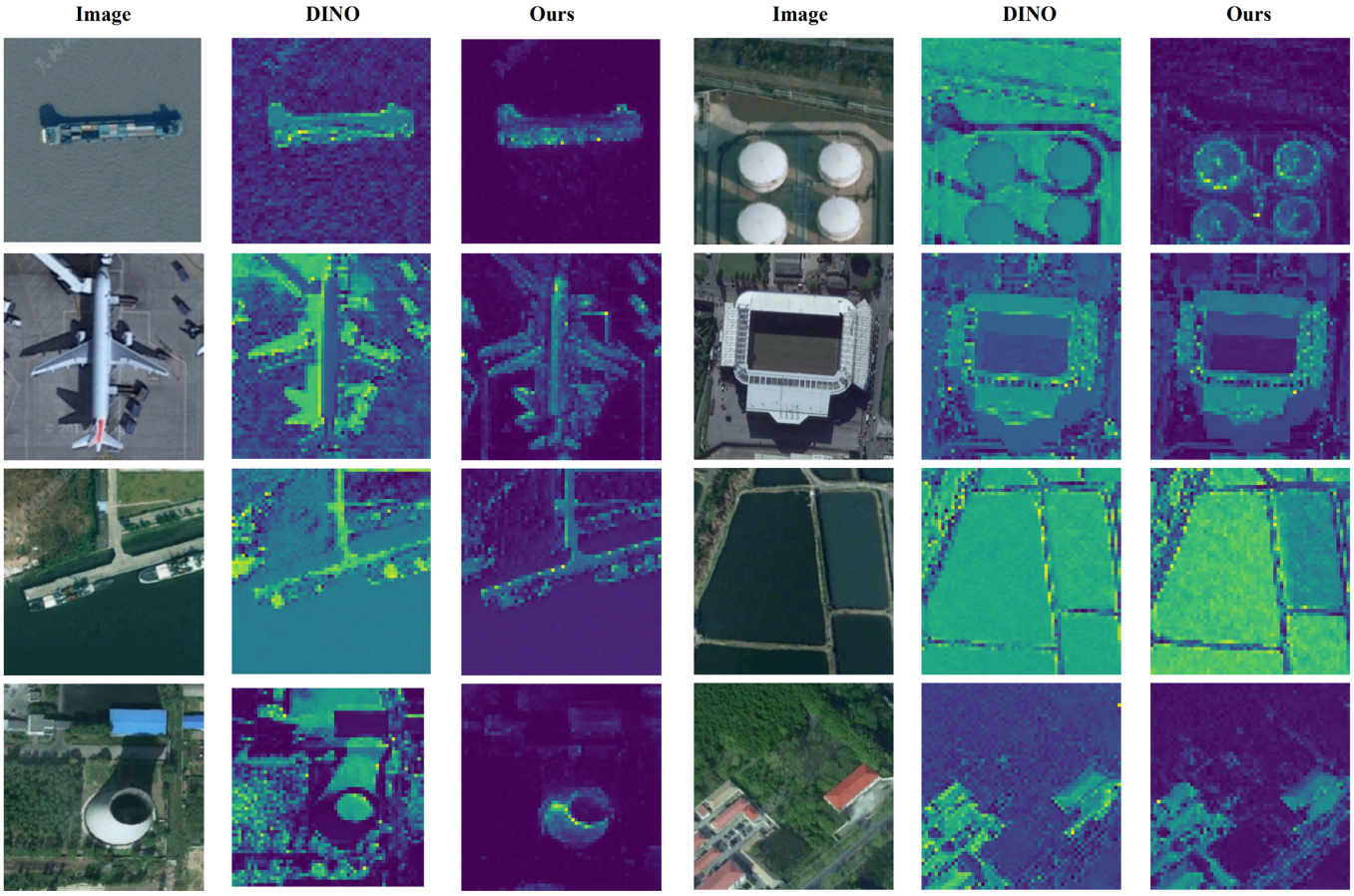


Fig. 9. Visualization results of the self-attention mechanism for the [CLS] token.

TABLE III
EFFECTIVE ABLATION EXPERIMENTS FOR EACH MODULE IN THE METHODOLOGY

	CPI		CRS	NWPU-RESISC45	LoveDA	
	Migration Aug	Do		(0A)	(mIoU ,mAcc)	
PCIR(Ours)	×	×	×	77.04	36.77	47.24
	×	×	✓	77.36	37.01	48.03
	×	✓	×	77.21	36.98	47.50
	✓	✓	×	77.40	37.05	47.88
	✓	✓	✓	77.50	37.44	48.79

TABLE IV
COMPARATIVE RESULTS OF THE EFFECTIVENESS OF THE COMBINED STRATEGY (CPI AND CRS) ON DIFFERENT MODELS

Dataset and Metric		ReLIC		DINO	
		Without	With	Without	With
NWPU-RESISC45	OA	74.79	76.94	77.04	77.69
LoveDA	mIoU	34.77	36.77	34.77	37.08
	mAcc	45.24	48.02	45.24	48.41

on both datasets. This finding suggests that the bidirectional causal regularization constraints in our method have a stronger impact on representation similarity optimization than the contrastive loss alone. Although the enhancement effect of the CPI module is not as pronounced as that of the CRS module, it still offers positive feedback on representation recognition performance. This indicates that feature-level stylistic interventions on images enable the model to focus more explicitly on and learn from the feature content regions, thereby achieving better feature recognition and understanding.

Finally, we test all our proposed policy combination architectures on two self-supervised methods, and the before-and-after comparison results are shown in Table IV. The results show that on the NWPU-RESISC45 dataset, our strategy architecture has an average increase of 1.4% in OA before and after the two methods, and on the LoveDA dataset,

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it has average increases of 2.16% and 2.98% in the mIoU and mAcc indicators, respectively; this means that our strategy can be structured to further focus on invariant representations of images and perform separation learning in different self-supervised models.

V. Discussion

A. Key to Invariant Characterization of Remotely Sensed Images: Discovering Causal Relationships in Images

Remote sensing images have significant characteristics, such as multiple features, multiple mixtures, and multiple changes, which present great challenges for representation learning. The PCIR method proposed in this article opens a new path for remote sensing representation learning by constructing a semantic-level causal graph model and introducing a causal partitioning intervention strategy. Our approach is based on two core principles of causal reasoning: the principle of common cause and the principle of independent causal mechanisms. By modeling the process of remote sensing image generation, we clearly partition the content and style of remote sensing images at the semantic level and apply intervention operations to ensure that the content representations remain unchanged while revealing the diversity of the style representations. This explicit causal modeling, on the one hand, avoids complete dependence on the statistical correlations in the data and directly mines causally invariant representations that play a decisive role in the task; on the other hand, it compensates for the shortcomings of traditional representation learning methods, which insufficiently account for the complexity of feature characteristics. This not only improves the model's adaptability to diverse remote sensing scenarios but also significantly enhances the model's ability to focus on and interpret the core feature regions.

In terms of the experimental results, PCIR validates the advantages of representation learning from a causal perspective, mainly in the following aspects.

1) *The representations are more interpretable.* By formalizing the representation process in a causal diagram through an SCM from a causal perspective, the causal relationships between feature characteristics are clearly demonstrated. Following the causal reasoning mechanism, we perform a series of interventions, which makes both our model and learning outcomes more interpretable.
2) *Cross-domain generalization is improved for open tasks.* The core design of PCIR utilizes CI to establish a clear causal relationship between content and style representations. Through causal partitioning interventions, the effects of environmental bias and sensor interference are eliminated, enabling the model to learn invariant causal representations that remain stable across different data domains. This representation not only shows strong robustness within a single domain but also transfers efficiently between domains. Furthermore, attention visualization results show that PCIR consistently focuses on the core feature regions, with the model's attention distribution remaining centered on key semantic areas regardless of background style variations. This stability verifies that causal representation learning captures consistent core features across domains.

B. Rethink and Future Work

Although the PCIR method demonstrates significant advantages in current representation learning, there are still some limitations that need improvement. First, in the partition intervention module, we use object detection labels for region segmentation, which makes the method highly dependent on high-quality, fine-grained labeled data, thereby limiting its applicability to weakly labeled or unlabeled datasets. To reduce this dependence on labeled samples, future research can actively explore integration with the rapidly developing large model technologies. For example, large zero-shot detection models such as SAM [63], [64], [65] could be introduced. With their strong generalization ability and excellent zero-shot segmentation performance, these models can directly partition semantic regions, thereby significantly reducing—or even eliminating—the dependence of PCIR on traditional labeled samples during implementation. Second, although this article designs and simulates three types of offsets—namely, the intrinsic offset, radiometric offset, and sensor offset of target ground objects—to achieve style enhancement for remote sensing images, we acknowledge that real-world variations in remote sensing imagery are usually far more complex. These may involve structural deformations, seasonal vegetation changes, and diverse acquisition resolutions. Future research will focus on designing broader and more realistic offset simulation strategies. In particular, with the increasing maturity of modern image generation technologies such as diffusion models and their growing application in the field of remote sensing, we can leverage these advanced models to simulate more complex and diverse data offset mechanisms, and even construct synthetic offset datasets that reflect multisource heterogeneity and complex environmental variations. This will help to further enhance the robustness and generalization ability of the model in open-world scenarios.

VI. Conclusion

In this article, we propose a new approach to remote sensing image representation learning from a causal perspective, termed PCIR. Unlike traditional global causal models, PCIR incorporates a semantic-level causal graph designed to capture the multifeature, multimixing, and multivariate characteristics of remote sensing imagery. Through causal partitioning interventions and symmetric causal regularization, PCIR effectively separates content-invariant representations from stylistic variations induced by environmental factors. Across two tasks and four datasets, our method demonstrates strong robustness and generalization capability. In the scene categorization task, PCIR achieves an average OA improvement of 1.79%, while in the semantic segmentation task, mAcc improves by up to 3.55%. The visualization of feature representation regions further shows that the model's attention aligns closely with human perceptual cognition, accurately capturing stable and semantically meaningful invariant features in remote sensing images. This not only enhances the model's reliability in complex remote sensing scenarios but also improves the interpretability of its behavior. Ablation experiments further

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confirm the relative independence and effectiveness of the causal partitioning intervention and symmetric causal regularization modules, laying a solid foundation for the overall performance improvement of PCIR. Our results highlight the importance of incorporating semantic-level causality in remote sensing image analysis and verify the feasibility and effectiveness of partitioned causal modeling in remote sensing image interpretation. This approach not only improves recognition accuracy but also provides a new direction for causal representation learning in dynamic and complex environments. In future work, we will focus on optimizing the causal partitioning intervention strategy and exploring its application to more fine-grained tasks, further enhancing the adaptability and scalability of causal models in remote sensing.

APPENDIX

A. Equivalence Derivation

Executing the $\mathrm{do}(S = s_i)$ operation in the causal graph signifies externally forcing the style variable S to a specific value s_i, thereby severing its natural generation mechanism from other variables in the model. Since we assume C and style S are independent, intervening on S does not alter the original distribution of content $P(C)$. Consequently, the distribution of images E after the intervention $\mathrm{do}(S = s_i)$ can be expressed as

$$P_{\mathrm{do}(S=s_i)}(E) = \int P(E \mid C = c, S = s_i) P(C = c)\, dc. \quad (15)$$

By integrating over all content C sampled according to its original distribution, $P_{\mathrm{do}(S=s_i)}(E)$ precisely describes the image distribution where "content follows the original distribution $P(C)$, and style is fixed at s_i."

Ideally, we assume that the data augmentation operation a_t possesses the characteristic of style substitution while preserving content *invariance*. This implies that if an original image E is generated from a specific content c and an original style s_{orig} [i.e., $E = f(c, s_{\mathrm{orig}})$], then the augmented image $E' = \mathrm{do}_{a_t}(E)$ should effectively replace the style with the target style s_i while maintaining the same content c. Formally, this can be expressed as

$$E' = \mathrm{do}_{a_t}\left(f\left(c, s_{\mathrm{orig}}\right)\right) \approx f(c, s_i). \quad (16)$$

Under this ideal assumption, the distribution of the augmented image E' can be obtained by marginalizing over content

$$P\left(E'\right) = \int P\left(E' \mid C = c\right) P(C = c)\, dc. \quad (17)$$

We can approximate $P(E' \mid C = c)$ as $P(E \mid C = c, S = s_i)$. Thus, the distribution of the augmented image E' can be rewritten as

$$P\left(E'\right) \approx \int P(E \mid C = c, S = s_i) P(C = c)\, dc. \quad (18)$$

By comparing the expressions, we can conclude

$$P_{\mathrm{do}(S=s_i)}(E) \approx P\left(E'\right) = P\left(\mathrm{do}_{a_t}\right)(E). \quad (19)$$

According to the linear relationship of X, Y, and E in the causal diagram model, the above equation holds equivalently for the three.

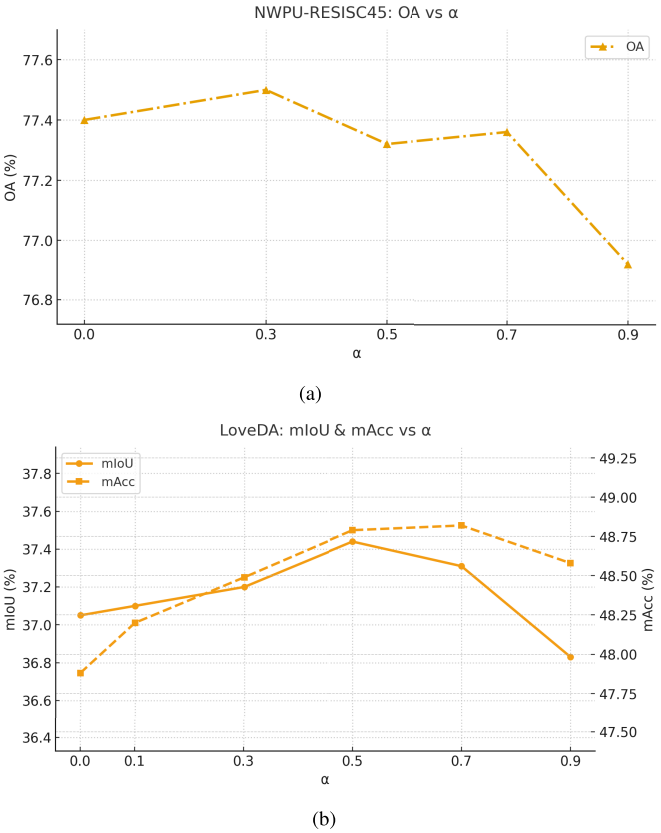


Fig. 10. Sensitivity analysis results of the weight parameters show that different tasks may have different optimal parameter values. (a) Scene classification: NWPU-RESISC45. (b) Semantic segmentation: AID dataset.

B. Hyperparameter Analysis

To verify the influence of the core hyperparameter α on the model performance and decoupling effect, we conducted a detailed sensitivity analysis. α serves as the weight of the penalty term in our overall optimization objective, aiming to guide the model to decouple causal representations. We believe that the value range of α is limited within $[0, 1]$, which is based on the principle of "main task-driven and regularization term assisted," to ensure that α only provides fine and auxiliary optimization constraints, avoiding its excessive weight from disrupting the model's learning of core content features, thereby ensuring the stability of training and the effectiveness of decoupling. We conducted tests within the range of $\alpha \in \{0.0, 0.3, 0.5, 0.7, 0.9\}$, and the experimental results are presented in Fig. 10.

First, when $\alpha = 0.0$ (i.e., there is no penalty term), the performance of all tasks is at its lowest level. This strongly demonstrates the necessity of explicitly introducing a penalty term guided by α for enhancing the invariance of representations. Second, the chart shows that the optimal α values for different tasks vary slightly. For instance, the peak of the NWPU task occurs at 0.3, while the peak of the mIoU/mAcc of LoveDA appears at 0.5/0.7. This multimodal phenomenon is reasonable because it demonstrates that different cross-domain migration tasks have different domain gaps, and thus, the decoupling strength requirements imposed on the penalty term are also different. Given that our goal is to find a universal and

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robust hyperparameter, we have observed that values around $\alpha = 0.5$ can maintain relatively optimal average performance and the most stable high-level performance across all critical mission metrics. Therefore, in most tasks, we can set α to 0.5. If we want to achieve the best performance, we can make adjustments for different tasks.

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